BLIND DETECTION OF ECLOSION FORGERIES BASED ON CURVELET IMAGE ENHANCEMENT AND EDGE DETECTION

YIN-CHENG QI
Department of Electronic and Communication Engineering
North China Electric Power University
Baoding, China
e-mail: qiych@126.com

XIAO-SHUANG XING
School of Electronics and Information Engineering
Beijing Jiaotong University
Beijing, China
e-mail: doublexiaoshuang@163.com

HUA-FANG-ZI ZHANG
Department of Electronic and Communication Engineering
North China Electric Power University
Baoding, China
e-mail: im.is@163.com

***Abstract*—Digital image forgery is a growing problem as the image could be easily manipulated. Digital image forgery detection has recently received significant attention. A fast and efficient blind detection algorithm is presented for the detection of eclosion forgeries. The second generation of Curvelet transform is firstly used to enhance the image, and then Canny operators is used to detect the edge of the enhanced image, finally the eclosion forgeries can be identified from the edge image. Experimental results show that the algorithm has less operation time and can distinguish the weak edges from the eclosion edges effectively. So it has a higher detection precision and efficiency.**

Key words-eclosion forgery; Curvelet transform; image enhancement; edge detection

I. INTRODUCTION

In 21st century, the rapid development of digital cameras and other digital imaging equipment provide a convenient way for obtaining digital images, the proliferation of computers makes it possible to store a large number of digital images, and the number of digital images is rapidly increasing. At the same time, users can conveniently manipulate and alter digital photographs by utilizing Photoshop and other software. In most cases, digital forgeries usually leaving no visual clues of tampering can be undistinguishable from authentic ones through our naked eyes, and therefore, the tampered photographs if used in news, academic research, political activities, legal events, will become incredible. So the research on the authenticity and integrity of digital images for digital image forensics is undoubtedly of great significance.

Currently, there are two types of digital image forensics methods: active and passive blind forensics. Active forensics is to rely on evidence of pre-embedded digital image watermark or digital signature to verify the authenticity of the image. Passive blind forensics is a digital image forensics technology independent of any pre-embedded information. This paper studies the edge feathering for image tampering forensics algorithm.

Edge feathering is a fuzzy processing on the blunt edge after a new image blended into an old one. This makes the transition gradual and therefore less noticeable.

In recent years, researchers proposed some tamper detection algorithms for copy-paste processing. J. Fridrich et al proposed an approximate block-matching based copy-move forgery detection algorithm in digital image[1]; Alin and Farid proposed a duplicated regions detection method by applying the principal component analysis and lexicographically sorting [2]; Myna et al proposed a duplicated regions detection algorithm using wavelets and log-polar mapping [3]; Bayram et al proposed a detection algorithm to use the Fourier-Mellin transform feature vectors and the notion of counting bloom filters as an alternative to lexicographic sorting [4]; Zhang et al proposed an image based on pixel matching copy-move tamper detection algorithm [5]. These algorithms can only detect the copy-paste tamper in the same image, but not in multiple images tampering.

The copy-move tampering between multiple images often will leave some hard edges. Now a lot of image editing software can feather the edges on the image, blurring the hard edges, leaving no visual cues of tampering. Focus on this kind of tampering, Hsiao and Pei proposed to use knowledge of domain and fuzzy estimation to detect fuzzy tampering [6]; Chen *et al* proposed a boundary smooth tampered detection algorithm to use the inharmonic points estimation (IPE) algorithm. Smoothing would reduce inconsistency among pixels and original inharmonic points may become moderate and the disharmony may disappear after smoothing. So if most pixels in a region are inharmonic points, it is reasonable to hold that the region has not been smoothed. In the same way, a smoothed region can be identified when most inharmonic points in the region vanish obviously [7]. Zhao proposed a method to detect the tampered region by image decomposition and profile reconstruction. If a region in image should be profile or edge, but there is no profile or edge in the reconstructed image at the correspondent region, this region should be tampered [8]. Blind detection algorithm of edge eclosion forgeries is still in development. A fast and efficient blind detection algorithm of edge eclosion forgeries is proposed in this paper. By using

978-0-7695-4356-7/11 $26.00 © 2011 IEEE
DOI 10.1109/CMSP.2011.70

Authorized licensed use limited to: University of Zagreb: Faculty of Electrical Engineering and Computing. Downloaded on March 24,2026 at 11:46:55 UTC from IEEE Xplore. Restrictions apply.

Curvelet edge enhancement, the proposed algorithm can identify weak edges and feathering edges, the detection accuracy is improved.

II. THE PRINCIPLE OF ALGORITHM

After feathering the edges of the tampered image, feathered edges will become blurred. Edge detection operation at the feathered edges will not form the edge image. Based on this characteristic [8], a blind detection of eclosion forgeries is proposed. Because there are some weak edges in natural images, these weak edges are not easily detected by edge detection method, which can easily be mistaken for tampering feathered edge. In order to distinguish the weak edges from feathering tampering and improve the detection accuracy, we enhance the edge of the test image before the edge detection. It will enhance the weak edges in the test image and doesn't affect the feathering regions. Simulation results show the effectiveness of the method.

III. CURVELET BASED IMAGE ENHANCEMENT

A. Curvelet transform

Curvelets as proposed by E. Candès and D. Donoho, constitute a relatively new family of frames that are designed to represent edges and other singularities along curves much more efficiently than the traditional wavelet based transforms. The idea of Curvelets is to represent a curve as a superposition of functions of various lengths and widths obeying the scaling law width ≈ length^2. This can be done by first decomposing the image into subbands, then, each scale is analyzed by means of a local Ridgelet transform, which makes the realization of the first generation of Curvelet transform is very complex and the decomposition of the pyramid. Curvelet also brought a lot of redundant data. Therefore, Candès et al proposed a second generation of Curvelet transform, the Curvelet base is given directly in the frequency domain, and a fast discrete realization algorithm is given, too. The second generation Curvelet transform has nothing to do with the Ridgelet theory. Its implementation process is simpler and faster. There are two ways, namely USFFT algorithm and Wrapping algorithm. Wrapping algorithm is the fastest Curvelet transform. For an N×N pixels image, its computational complexity is O ($N^2 \log N$), only 6 to 10 times of fast Fourier transform. Detailed realization procedure of Wrapping algorithm is:

a) 2D FFT： Apply the 2D FFT to the two-dimensional function $f(t_1, t_2)$ in $L^2(R^2)$ and obtain Fourier samples $\hat{f}[n_1, n_2], \quad -n/2 \le n_1, n_2 \le n/2$.

b) Resample: In the frequency domain, for each scale/angle pair (j, l), resample (or interpolate) $\hat{f}[n_1, n_2]$ to obtain sampled values $\hat{f}[n_1, n_2 - n_1 \tan\theta_l]$, where, θ_l is the rotation angle sequence.

c) Localization: multiply the interpolated object $\hat{f}$ with window function $\tilde{U}_j$ and obtain the localized $\hat{f}[n_1, n_2]$.

$$\hat{f}[n_1, n_2] = \hat{f}[n_1, n_2 - n_1 \tan\theta_l]\tilde{U}_j[n_1, n_2] \tag{1}$$

d) Wrapping: Wrap this product around the origin and obtain $\hat{f}_{j,l}$.

e) inverse 2D FFT: apply the inverse 2D FFT to each $\hat{f}_{j,l}$, hence collecting the discrete coefficients $c^D(j,l,k)$.

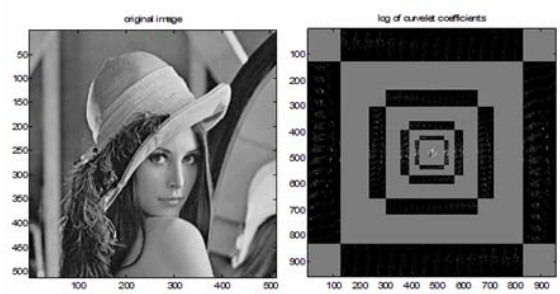
B. Curvelet coefficients

After Curvelet transform, the structure of Curvelet coefficients $C\{j\}\{l\}(k_1, k_2)$ for 512×512 pixels image is shown in Table I, where, j is scale factor, l denotes direction, $k=(k_1, k_2)\in Z^2$ denotes the position coordinates of the coefficients. After Curvelet transform, a 512×512 pixels image is divided into six scale layers. The first layer is the coarse scale, is a 32×32 matrix of low-frequency coefficients, reflecting the profile of the image. The sixth layer is the finest scale, which is a 512×512 matrix composed of the high frequency coefficient, reflecting the image details and the edge features. The second layer to the fifth layer as the fine scale layers, each layer is divided into four directions, each direction is divided into 8, 8, 16, 16 small direction. Each small direction is a matrix composed of high-frequency coefficients. The edge features of these layers have more directions.

Fig. 1 (a) is the original image Lenna. Fig. 1 (b) is the Curvelet transform coefficients of Lenna.

TABLE I. STRUCTURE OF CURVELET TRANSFORM COEFFICIENTS

Layer	Scale j	Direction l	Matrix				
Coarse	C{1}	1	32×32				
Fine	C{2}	32 (4×8)		16×12	12×16	16×12	12×16
	C{3}	32 (4×8)		32×22	22×32	32×22	22×32
	C{4}	64 (4×16)		64×22	22×64	64×22	22×64
	C{5}	64 (4×16)		128×44	44×128	128×44	44×128
Finest	C{6}	1	512×512				



(a) (b)

Figure 1. Image of Curvelet transform coefficients: (a) is original image Lenna, (b) is Curvelet transform coefficients.

Authorized licensed use limited to: University of Zagreb: Faculty of Electrical Engineering and Computing. Downloaded on March 24,2026 at 11:46:55 UTC from IEEE Xplore. Restrictions apply.

Apply the inverse Curvelet transform to the coefficients of the first layer (coefficients of other layers are set to zeros), we can get the reconstructed image only from this layer, and the reconstructed images of other layers are shown in Fig. 2.

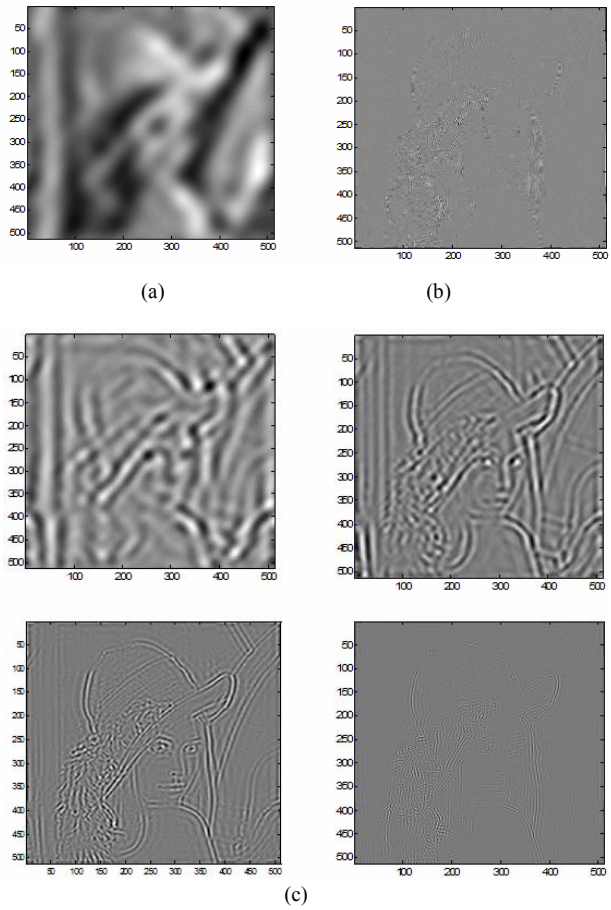


Figure 2. Reconstructed image of single layer Curvelet coefficients of the image Lenna: (a) coarse scale coefficients, (b) finest scale coefficients, (c) fine scale coefficients.

C. *Curvelet transform based image enhancement*

Curvelet transform can represent edges and other singularities along curves much more efficiently. In order to enhance the edge of original image, Curvelet transform coefficients must be modified. The main idea of Curvelet transform based image enhancement is:

Firstly, apply Curvelet transform to the image to be detected, and obtain the multi-scale Curvelet coefficients $C\{j\}\{l\}(k_1, k_2)$.

Then, the enhancement function $y_c(x)$ is used to process the Curvelet coefficients according to the noise level of each scale layer, to achieve piecewise nonlinear enhancement. The enhancement function $y_c(x)$ is described as follows:

$$\begin{cases} y_c(x) = 1 \quad x < T_{\min} \\ y_c(x) = \dfrac{x - T_{\min}}{T_{\min}}\left(\dfrac{T_{\max}}{T_{\min}}\right)^p + \dfrac{2T_{\min} - x}{T_{\min}} \quad x < 2T_{\min} \\ y_c(x) = \left(\dfrac{T_{\max}}{x}\right)^p \quad 2T_{\min} \le x < T_{\max} \\ y_c(x) = 1 \quad x \ge T_{\max} \end{cases} \tag{2}$$

where p denotes the nonlinearity, $T_{\min}$ and $T_{\max}$ are two constant thresholds. $T_{\min}$ is related to the noise level of Curvelet coefficient in each scale layer. $T_{\max}$ is related to the maximum M_c of Curvelet coefficient in correspondent subband, $T_{\max}=qM_c$, $q<1$.

Finally, apply the inverse Curvelet transform to processed transform coefficients to obtain the enhanced image. [10]

IV. CANNY EDGE DETECTION

In this paper, the traditional Canny edge detection algorithm is employed on the enhanced image to detect edges, the main steps are described as follows [11]:

a) Noise reduction: a Gaussian filter is used to smooth the image and reduce the noise. Calculating the convolution of the original image $i(x, y)$ and the two-dimensional Gaussian function in rows and columns, respectively, obtain the smoothed image $f(x, y)$. The two-dimensional Gaussian function is

$$G(x, y) = \frac{1}{2\pi\sigma^2} e^{-(x^2+y^2)/2\sigma^2} \cdot \tag{3}$$

b) Gradient calculation: after smoothing the image and eliminating the noise, the next step is to find the edge strength by taking the gradient of the smoothed image. The first order derivative operator is used to calculate the gradient magnitude and direction of the smoothed image at each point. Let G be the gradient magnitude image, and θ denote the gradient direction image.

c) Non-maximum suppression: after the edge directions are known, non-maximum suppression now is applied. Non-maximum suppression is used to trace along the edge in the edge direction and suppress any pixel value (sets it equal to 0) that is not considered to be an edge. This will give a thin line in the output image.

d) Hysteresis thresholding: finally, hysteresis is used as a means of eliminating streaking. Hysteresis uses 2 thresholds, a higher T_2 and a lower T_1, and $T_1=0.4\times T_2$. Any pixel in the image that has a value greater than T_2 is presumed to be an edge pixel, and is marked as such immediately. Then, any pixels that are connected to this edge pixel and that have a value greater than T_1 are also selected as edge pixels. If you think of following an edge, you need a gradient of T_2 to start but you don't stop till you hit a gradient below T_1.

V. DETECTION ALGORITHMS

The main steps of Curvelet image enhancement and Canny edge detection based eclosion forgeries detection method are:

Authorized licensed use limited to: University of Zagreb: Faculty of Electrical Engineering and Computing. Downloaded on March 24,2026 at 11:46:55 UTC from IEEE Xplore. Restrictions apply.

Step 1: input image data X.

Step 2: Curvelet transform based image enhancement on image X [10]. The Procedure is as follows.

a) Curvelet transform: apply Curvelet transform to image X and obtain Curvelet coefficients.

b) Noise level estimation: the noise level $\delta_{i,\ j}$ of Curvelet coefficient $C_{j,\ l}$ is estimate for scale j and angle l, T_{min} and T_{max} are calculated.

c) Curvelet coefficients enhancement: calculate enhancement function $y_c(|C_{j,\ l}|)$, and each Curvelet coefficient $C_{j,\ l}$ is multiplied by $y_c(|C_{j,\ l}|)$.

d) Inverse Curvelet transform: the inverse Curvelet transform is done on modified coefficients, the result is the enhanced image $X_{enhanced}$.

Step 3: Canny operator is applied to the enhanced image $X_{enhanced}$ and get the edge image.

VI. EXPERIMENTAL RESULTS

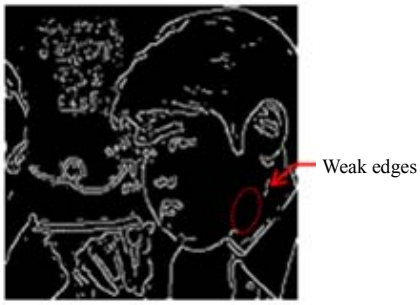
"Network chubby", the forged image, is used as the test image in this paper. Experimental results of the proposed method, inharmonic points estimation (IPE) algorithm, image decomposition and contour construction based algorithm are shown in Fig. 3.

Fig. 3(a) is a stage photo of the Yesterday Once More, where Andy Lau's face was altered into the face of "Network Chubby". Feathering operation is performed on the borderline between “Network Chubby’s” face and Andy Lau's neck and the edge is smoothed in this region. The proposed algorithm, IPE algorithm, image decomposition and contour construction based detection algorithm are simulated in the same computer. In order to save computing time, we only select a part of the original image (256×256 pixels) to verify the effects. Fig. 3(b) is the result of IPE algorithm. From Fig. 3(b), it can be seen that IPE algorithm can not accurately locate the tampered region. Fig. 3(c) and (d) are the result of image decomposition and contour construction based detection algorithm and the proposed algorithm respectively. From Fig. 3(c) and Fig. 3(d), it can be seen that the proposed algorithm and image decomposition and contour construction based algorithm can accurately locate the tampered region. Because in the region of elliptical dotted line, edge or contour should exist, while the edge (or contour) can not find in the experimental results. It means that this region is tampered by feathering. In addition, the location of the arrow pointed in Fig. 3 (d) is a weak edge and not a feathering region. Image decomposition and contour construction based algorithm convicted that this region is tampered too, while the proposed algorithm can identify weak edges and feathering edges. So the detection accuracy is improved. Simulation time of these three algorithms under the same conditions is 239.094s, 569.19s and 5.047s, respectively. It can be seen that the proposed algorithm significantly reduces the simulation time.

(a)

(b)

(c)



(d)

Figure 3. Detection results of three methods: (a) stage photo of Yesterday Once More, Andy Lau's face was altered into the face of "Network Chubby", (b) results of IPE algorithm (c) results of image decomposition and contour construction based algorithm, (d) results of the proposed method.

VII. CONCLUSIONS

As eclosion forgeries have become popular, the importance of forgery detection is increased recently. A fast and efficient blind detection method for eclosion forgeries is proposed in this paper. After a Curvelet transform based image enhancement, Canny edge detection operator is applied to the enhanced image and get the edge image. The tampered region can be determined by different edge images. Experimental results show that the proposed algorithm can accurately locate the tampered region, distinguish between the weak edges and the feathering edges of tampering. At the same time, simulation time of the proposed algorithm is less than other two methods.

Authorized licensed use limited to: University of Zagreb: Faculty of Electrical Engineering and Computing. Downloaded on March 24,2026 at 11:46:55 UTC from IEEE Xplore. Restrictions apply.

REFERENCES

[1] J. Fridrich, D. Soukal, and J. Lukas, "Detection of copy-move forgery in digital images," Proc. Digital Forensic Research Workshop, Cleveland, OH, August 2003.

[2] A.C. Popescu and H. Farid, "Exposing digital forgeries by detecting duplicated region," Technical Report, TR2004-515, Dartmouth College, Computer Science. http://www.cs.dartmouth.edu/ farid/ publications/ tr04.pdf.

[3] A. N. Myna, M. G. Venkateshmurthy, and C. G. Patil, "Detection of region duplication forgery in digital images using wavelets and log-polar mapping," Proc. IEEE Int'l Conf. on Computational Intelligence and Multimedia App., July, 2007, Vol. 3, pp. 371-377.

[4] S. Bayram, H. T. Sencar, and N. Memon, "An efficient and robust method for detection copy-move forgery," 2009 IEEE Int'l Conf. on Acoustics, Speech and Signal Processing, pp.1053-1056.

[5] J. Zhang, L. Zhan, and Y. Su, "An algorithm for image copy-paste forgery detection by pixel matching," Journal of Tianjin University, Vol. 42, No. 8, 2009, pp.714-720.

[6] D. Y. Hsiao, and S. C. Pei, "Detecting digital tampering by blur estimation," Proc. the First Int'l Workshop on Systematic Approaches to Digital Forensic Engineering, 2005, pp. 264-278.

[7] Y. Chen, and Y. Wang, "Exposing digital forgeries by detecting traces of smoothing," IEEE the 9th Int'l Conf. for Young Computer Scientists, 2008, pp. 1440-1445, doi: 10.1109/ ICYCS. 2008. 448.

[8] F. Zhao, X. Li, Y. Peng, Y. Huo, "Blind detection of digital forgeries using detection trace of eclosion," Signal Processing, Vol. 25, No.12, 2009, pp. 1922~1927.

[9] J. Yan, and X. Qu, Beyond Wavelets and its Applications, Beijing: National Defense Industry Press.

[10] Z. Zhao, Z. Geng, Y. Zhang, X. Sui, "Adaptive image enhancement algorithm based on second generation curvelet transform," Computer Engineering and Applications, Vol. 45, No. 9, 2009, pp. 192-195.

[11] T. Luo, X. Zheng, and T. Ding, "Improved self-adaptive threshold canny edge detection," Opto-electronic Engineering, Vol. 36, No. 11, 2009, pp. 106-111.

Authorized licensed use limited to: University of Zagreb: Faculty of Electrical Engineering and Computing. Downloaded on March 24,2026 at 11:46:55 UTC from IEEE Xplore. Restrictions apply.